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Dependence of Properties of Zirconium Nitride Nanoparticles Prepared by Laser Ablation in Liquids on Synthesis Parameters

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Abstract—This paper presents a study of the dependence of the size and optical characteristics, as well as the productivity of laser synthesis of zirconium nitride (ZrN) nanoparticles (NPs) on laser ablation parameters. It is shown that changing the distance between the focusing lens and the target surface allows one to control the mode and half-width of the size distribution from (59 ± 35 nm) to (29 ± 14 nm), as well as spectrally shift the plasmon resonance peak from 638 nm to 680 nm. Additionally, a possibility of controlling the size characteristics of ZrN NPs when changing the laser radiation energy is demonstrated. Increasing the laser pulse energy from 10 to 100 μJ makes it possible to vary the NP size from (27 ± 13 nm) to (47 ± 30 nm). The results form a quantitative basis for the production of laser-synthesized ZrN NPs with controlled size and optical properties for applications in biomedicine.

Keywords: zirconium nitride, nanoparticles, laser ablation in liquid, size, optical properties

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1. INTRODUCTION

Unique optical, mechanical, and thermal properties of nanomaterials based on nitrides of group IV transition metals (TiN, ZrN, and HfN) open up fundamentally new prospects for the development of methods of cancer theranostics [1]. Pronounced optical absorption in the transparency window of biological tissues (650–950 nm) determines the high potential of such nanoparticles (NPs) for sensitization of targeted photothermal therapy and photoacoustic imaging [1–3]. In addition, the relatively low work function and catalytic activity of ZrN make this material promising for the use in radiation and proton therapy [4, 5].

Another important property of nanomaterials for biomedical applications is related to their size characteristics. As an example, NPs with a size in the range of 30–60 nm have the highest efficiency of cellular internalization [6]. On the other hand, reducing the size of NPs leads to the increase of their circulation in the bloodstream and their better accumulation in the target area to be ensured. That is why the ability to control the size properties of NPs is an important task that forms a reliable foundation for future biomedical applications of nanomaterials.

Among the various methods of NP synthesis, laser ablation in liquids is an effective way to obtain ultrapure nanomaterials, providing precise size control without the need for complex chemical precursors and procedures [7, 8]. Pulsed laser ablation in liquids (PLAL) involves the removal of material from a solid target by the interaction of high-energy laser pulses with the target surface. PLAL is especially important for biomedical applications, since it allows NP contamination to be minimized during the synthesis, which is a critical factor when working with materials intended for the use in living organisms [9–11].

In this paper, we investigate the possibility of controlling the size and optical properties of ZrN NPs as a function of the synthesis parameters during PLAL. We determine a quantitative dependence of the half-

width of the size distribution, productivity, and optical properties of the NPs on the focusing parameters. An additional possibilitys for the control of size characteristics of ZrN NPs by varying the laser radiation energy is also demonstrated.

2. MATERIALS AND METHODS

Zirconium nitride NPs were synthesized by femtosecond (fs) PLAL. A ZrN target (99.99% purity, Girmet, Russia) was fixed vertically in a glass cell filled with 5 ml of acetonitrile, at a distance of 3 mm from the cell wall. A 3 mm diameter beam from a TETA 20 Yb:KGW laser (1030 nm, 250 fs, 10–100 μJ, and 100 kHz; Avesta, Russia) was focused by a flat-field objective (F-theta) with a focal length of 100 mm on a ZrN target surface through a side wall of the cell. The beam diameter on the target was ≈50 μm. To increase the efficiency of the synthesis, the laser beam moved along the target surface at a speed of 5 m/s using a galvanometric scanner (LScan-10, Ateco-TM, Russia). The duration of the synthesis procedure was 3 min.

The size and morphological properties of ZrN NPs were characterized using a MAIA 3 scanning electron microscope (SEM) (Tescan, Czech Republic). To calculate the size distribution, SEM images were processed in the ImageJ program. Optical extinction spectra were measured in the spectral range of 330–1100 nm on an ML122 spectrophotometer (SOL Instruments, Belarus). The hydrodynamic size distribution was measured using an autocorrelation spectrometer (Zetasizer ZS, Malvern Instruments, France) by the dynamic light scattering (DLS) method.

3. RESULTS AND DISCUSSION

In this work, the dependence of the nanoparticle properties on the lens position relative to the target surface was determined by an optical method and gravimetric measurements of the mass of dried NPs. Extinction spectra of colloidal solutions of ZrN NPs obtained at different distances from the lens to the target were measured (Fig. 1a). The position of the focal plane on the target surface corresponds to the highest extinction of NPs in the range of 330–350 nm, where the greatest contribution to the extinction is made by interband transitions in ZiN. In this case, a shift in the plasmon resonance peak in the range of 638–680 nm was observed with a change in the relative position of the lens from –5 mm to +3 mm. The zero point corresponded to the position of maximum efficiency of laser synthesis. Since at a distance of +5 mm, ablation of the glass cell surface occurred, this point was excluded from the measurements. The concentration of nanoparticles varied depending on the position of the lens relative to the target surface. In addition, gravimetric measurements of the NP synthesis productivity were carried out (Fig. 1b). It was found that the displacement of the target from the focal plane leads to a decrease in the synthesis productivity from 9.0 to 1.1 mg/h. In these experiments, the radiation energy was 20 μJ, the beam width at the waist was 50 μm, and the repetition rate was 100 kHz.

Changing the distance between the lens and the target also affected the size (hydrodynamic diameter) of the resulting NPs. The mode and half-width of the size distribution decreased with increasing distance from 59 ± 35 to 29 ± 14 nm (Fig. 2a). This dependence of the size properties of the synthesized nanoparticles can be explained by the fact that the focused laser radiation causes an optical breakdown in the liquid in front of the target when the distance from the lens to the target increases beyond the focal length. Thus, a part of the laser energy is absorbed and scattered by the plasma formed during the optical breakdown. As a result, a smaller fraction of the energy reaches the target surface, which, as shown below, leads to a decrease in the size distribution of the resulting NPs.

At the same time, the size characteristics of the ZrN NPs also tended to decrease with decreasing laser pulse energy (Fig. 2b). Changing the radiation energy from 10 to 100 μJ made it possible to control the mode and half-width of the distribution from 27 ± 13 to 47 ± 30 nm, respectively.

Additionally, the dependence of the size characteristics of the resulting NPs on the focusing parameters was studied using a SEM visualization (Fig. 3). It was found that all synthesized NPs have a close-to-spherical morphology. The obtained data have a similar trend with the results of DLS measurements. Zirconium nitride exhibits a nature close to metallic (semi-metallic) structure in the UV, visible, and near IR ranges, which explains the presence of plasmon resonance in the obtained NPs [12]. In this work, the shift of the plasmon resonance peak was caused by a change in the size distribution of the studied NPs (shift in the mode and change in the half-width).

Redispersion of NPs from organic solvents into an aqueous medium can only slightly affect their colloidal stability, while replacing the liquid medium with a biologically relevant system (e.g., phosphate buffered saline solution) often leads to a sharp decrease in colloidal stability and agglomeration due to the

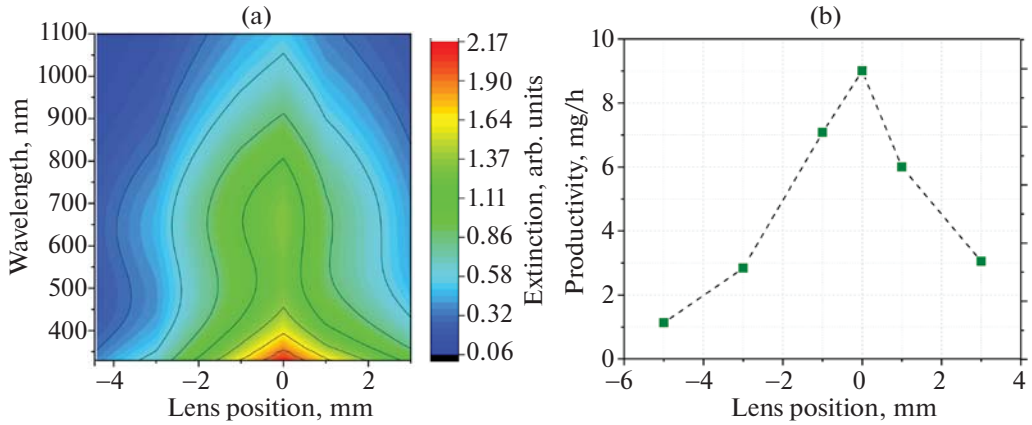

Fig. 1. (a) Extinction spectra of colloidal solutions of ZrN NPs as a function of the lens position relative to the target surface, and (b) laser synthesis efficiency as a function of the lens position relative to the target surface. The position of maximum laser synthesis efficiency corresponds to the zero point.

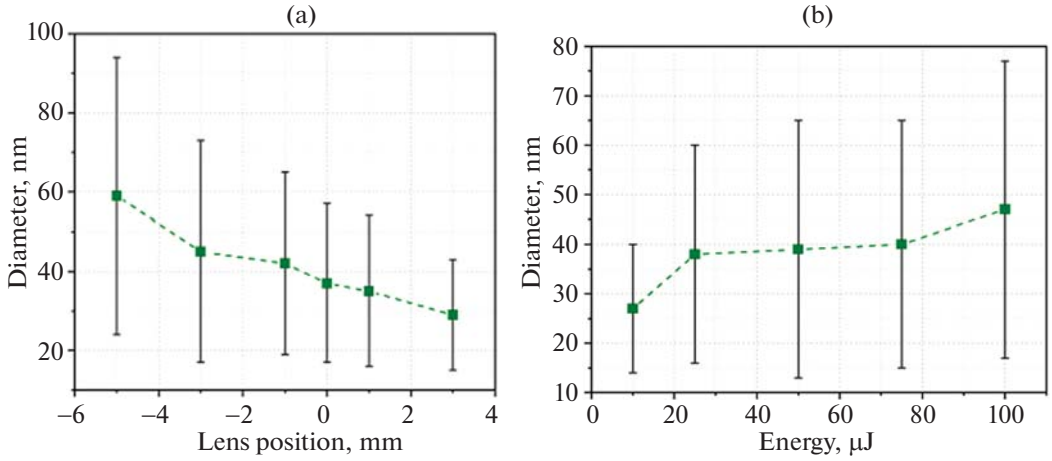

Fig. 2. (a) Mode (green squares) and half-width of the size distribution of ZrN NPs (vertical lines) as a function of the lens position relative to the target surface, and (b) mode and half-width of the size distribution of ZrN NPs as a function of the laser pulse energy. The size data are obtained using DLS. The position of maximum laser synthesis efficiency corresponds to the zero point.

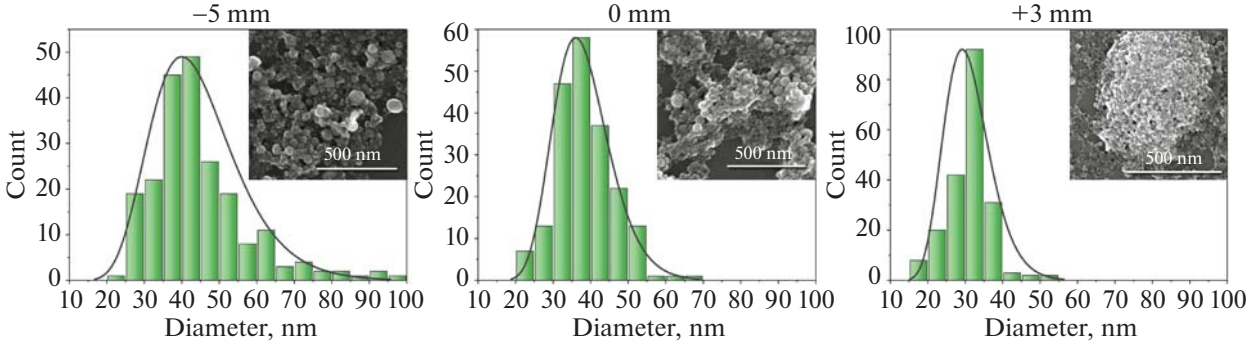

Fig 3. Size distribution and SEM images of ZrN NPs for different distances from the focusing lens to the target surface.

high ionic strength of such solutions. In addition, long-term incubation of NPs from noninert materials in such solutions can lead to a partial oxidation of nanomaterials. However, a surface modification of NPs using biocompatible polymer compounds (e.g., polyethylene glycol, polyacrylic acid, etc.) provides increased colloidal stability even in biologically relevant environments due to steric repulsion forces, and also acts as a protective inert layer from the effects of the surrounding liquid.

4. CONCLUSIONS

We have determined the dependence of the size and optical characteristics of laser-synthesized ZrN NPs on the synthesis parameters, and have shown that increasing the distance from the focusing lens to the target surface allows the average size of NPs to be decreased from 59 to 29 nm, with a change in the half-width of the size distribution from 35 to 14 nm, respectively. In addition, under the current conditions, we have also observed a shift in the plasmon resonance peak from 638 to 680 nm to the region of relative transparency of biological tissues. At the same time, the maximum productivity at 20 μJ and 100 kHz has reached 9.0 mg/h (25.0 fg/pulse) and decreased with distance from it to 1.1 mg/h (3.1 fg/pulse). However, varying the focusing parameters does not affect the morphology of the resulting NPs. All the obtained nanomaterials have demonstrated a close-to-spherical morphology. Changing the laser radiation energy also has made it possible to control the size properties of ZrN NPs. A decrease in energy from 100 to 10 μJ causes a decrease in the mode and half-width of the size distribution from 47 ± 30 to 27 ± 13 nm. The results form an experimental basis for the development of ultra-pure laser-synthesized ZrN NPs with controlled size and optical properties for applications in biomedicine.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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